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On Dissimilatory *r*-Loss in Greek

Brent Vine

ABSTRACT. Since Wackernagel, Hom. ἀργός 'bright; swift' has been assumed to arise from earlier *ἀργρός by dissimilatory *r*-loss, comparable to Ved. *ǰjrá-* 'id.' (cf. ἀργι- : Ved. *ǰji-*, etc.). More recent analyses deriving ἀργός from **h₂erǧ-o-* or **h₂ǰǧ-ó-*, argued on morphological grounds, have been largely ignored. The problem can be evaluated phonologically — and indeed, traditional treatments of Greek dissimilatory *r*-loss are unsatisfactory. Contemporary approaches to the phonology of dissimilation provide insight into Greek dissimilatory *r*-loss, and shed light on Wackernagel's "*ἀργρός" and on a series of problematic forms in Greek (γαστήρ, Myc. *re-u-ko-ro-o-pu₂-ru*, etc.) and PIE (**ti-sr-* '3 [f.]' < ***tri-sr-*).

1.1. One of the most enduring and pervasive dicta of Greek and Indo-European historical/comparative grammar, based on an insight of Wackernagel (1897:9), is the claim — now all but universally accepted — that Gk. ἀργός 'bright; swift' [Hom.+] evolved as a result of dissimilatory *r*-loss from an earlier *ἀργρός, and that this form, in turn, is directly comparable to Ved. *ǰjrá-* 'id.' and its Avestan cognate *ərəzra-*. This equation then helps validate this root (PIE **h₂erǧ-*) as a participant in the so-called "Caland system": cf. typical Caland alternations of the sort *ἀργρός, Ved. *ǰjrá-* beside ἀργι- : Ved. *ǰji-* (first-compound member), ἀργυ- : Ved. *ǰju-* (: TB *ārkwi* 'white'), etc.

This traditional analysis of ἀργός is a staple not only of the Greek etymological dictionaries (see GEW and DELG s.v.), including recent updates of DELG (e.g. Le Feuvre 2005), but also of virtually all standard handbooks of Greek historical grammar,[1] as well as handbooks of Indo-European linguistics,[2] and countless specialized studies of Greek phonology and morphology.[3] An exhaustive listing of secondary literature is neither possible nor necessary for such a widely-accepted feature of Greek and Indo-European historical grammar.

1.2. Yet despite this general agreement, a suggestion has been placed on the record by J. Rau (1998:159) that Gk. ἀργός could (and perhaps should) be taken at face value, i.e. as reflecting a full-grade thematic **h₂erǧ-o-*, a form that would be directly comparable to **h₁reud^{h}-o-* 'red' (cf. Go. *rauþs*, [non-]Lat. *rūfus*), from a root that has well-established associations with the Caland system. A similar perspective has emerged in work by A. Nussbaum (e.g. 1996), who operates with the other face-value interpretation of the form, i.e. via a zero grade **h₂ǰǧ-ó-*.[4] Given the entrenched status of "ἀργός < *ἀργρός", it is hardly surprising that this alternative conception of ἀργός has not yet found much favor, apart from the occasional brief notice.[5]

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1 E.g. Schwyzer 1939:447, Rix 1992:18, Meier-Brügger 1992:II.32.

2 E.g. IEW 64, Mayrhofer 1986:139, Beekes 1995:170, Fortson 2004:121 and 2010:135.

3 A few recent samples: Meissner 1998:238, Hackstein 2002:213, Probert 2006:160, Remmer 2007:129.

4 In both cases, the morphological argumentation is based in part on the denominative *t*-stem ** h₂(e)rǧ-é-t-* seen in Hom. obl. ἀργέτ- 'white'.

5 Thus e.g. Meier-Brügger 2003:288 (with both **h₂ǰǧ-ró-* and **h₂ǰǧ-ó-*), vs. Meier-Brügger 2000:271 (**h₂ǰǧ-ró-* only).

1.3. It should be noted, however, that the Rau-Nussbaum conception has so far been argued entirely on morphological grounds, which are after all relatively easy to dismiss, since the form is phonologically ambiguous even between full grade and zero grade. But in principle, the case can be evaluated from a phonological perspective — that is, from the standpoint of patterns of dissimilatory *r*-loss in Greek and the typology of such patterning elsewhere. As it happens, traditional treatments of the Greek side of the problem leave much to be desired. My immediate goal, then, will be to develop a more satisfactory account of dissimilatory *r*-loss in Greek. To do this, it will be useful to consider certain concepts that have become central to modern phonological theory in general, as well as some specific points arising from modern phonological studies of dissimilation. On this basis, it may then be possible to evaluate the ἀργός case from the perspectives of both Greek phonology and modern phonological theory.

2.1. The most thorough and explicit discussion of dissimilatory *r*-deletion in Greek, despite its brevity, appears in M. Lejeune's classic handbook of Greek phonology (1972:150). Lejeune observes, first, that Greek words with two instances of surface /r/ are phonetically well-formed and well-attested, for which he cites the six examples βάρβαρος 'foreign', γαργαίρω 'swarm', ἀραρίσκω 'join', ὄρωρα 'arose', πρῷρα 'prow', and τρήρων 'fearful'. Nevertheless, he continues, dissimilation "often" leads to the deletion of either the first /r/ or the second /r/: thus the first /r/ is deleted in Arg., Arc. φᾶτρᾱ 'phratry' (vis-à-vis Hom. φρήτρη, Att. φράτρα) and Phoc. φᾶτριᾱ 'id.' (cf. Att. φρᾱτρίᾱ); and the second /r/ is deleted in Cypr. *ve-re-ta* (= ϝρητᾱ) 'declaration' (cf. Att. ῥήτρᾱ, El. ϝρᾶτρᾱ), and in the background of δρύφακτος 'railing' (< *δρύ-φρακτος, cf. φραγῆναι 'fence in') and of μάρτῡς 'witness' (< *μάρτυρ-ς). This is the full extent of Lejeune's treatment.

2.2. Little of substance has been added to this picture in subsequent discussions, which merely mention the process, sometimes providing additional data (which will concern us below), and often with a nod in the direction of Lejeune. Thus the contemporaneous treatment by Poultney (1972) incorporates a few more forms, but in a very unsystematic manner; Threatte, in the context of Attic epigraphy (1980:481ff.), is especially concerned with distinguishing "true dissimilation" from careless engraving errors; Perpillou ([1984]1996:147f.) discusses a possible Mycenaean example; Blanc (1999:320f.) essentially repeats Lejeune, with a comment on reduplicated forms; Hackstein (2002:213) cites several standard examples and notes the independent nature of Greek and Indic *r*-dissimilation; and Tremblay (2003:30ff.) offers further discussion of Perpillou's case, with related material.

3.1. It is at this point that we can begin to engage with the theory and practice of modern phonology, by stating that Lejeune's treatment of dissimilatory *r*-loss does not meet even the most minimal requirements of a phonological analysis, as these are understood today. In essence, Lejeune's approach is only slightly removed from an older descriptive tradition typified by such classic studies of dissimilation as those by Grammont (1895, 1933:269ff.). More recent approaches to phonological patterning — whether in the "structuralist" mode of the 1940's and 1950's, the "generativist" approach of the 1960's and 1970's, or various more recent "post-generative" frameworks — aim to "capture linguistic generalizations" that emerge from precise statements about phonological context and distribution, and to express these generalizations in the form of phonological rules (or constraints), insofar as possible, even for phenomena which, as in the case of dissimilation, are often described as "sporadic" or "optional".

3.2. What is particularly unsatisfying is that Lejeune takes no notice of the differences in phonological environment — surely significant — between the examples of double-*r retention* and the examples he provides of dissimilatory *r-loss*. In fact, the cases of *r*-loss he offers show two (and only two) highly restricted phonological contexts: first of all, consecutive syllables with identical onset clusters /Cr-/, in which either the first syllable-onset or the second is susceptible to *r*-loss, as in φᾶτρᾱ (with *r*-loss in the first syllable: /$Cr_1V.Cr_2V$-/ > [CøV.Cr_2V-]) or δρύφακτος (with *r*-loss in the second syllable: /$Cr_1V.Cr_2V$-/ > [Cr_1V.CøV-]); and secondly, represented by the single example μάρτῠς, consecutive syllables with coda-final /-r/ (and in this example, the second coda is the one that undergoes *r*-loss). Whereas all but one of Lejeune's examples of *r*-loss belong to the first type, this phonological environment does not occur in a single one of Lejeune's examples of *r*-retention (2.1.). Thus βάρβαρος has /r/ in non-consecutive syllables; ἀραρίσκω and ὄρωρα do have /r/ in the onsets of consecutive syllables, but show simple onsets /r-/ and not complex onsets /Cr-/; and while πρῶρα and τρήρων have /r/ in the onsets of consecutive syllables, in both cases the first appears in a cluster /Cr-/ and the second is a single /r-/, again failing to match the environment actually attested in the majority of Lejeune's cases of *r*-loss.

3.3. The only overlap with the forms showing retention of double /r/ involves μάρτῠς, which at first glance appears to coincide with the consecutive coda environment of γαργαίρω, at least in its immediate preform */gar.gar.i̯ō/ (and so also for other reduplicated intensives of this type: μαρμαίρω, πορφύρω, etc.). But there are several differences between the phonological (and morphological) environments of μάρτῠς (or rather its preform */mar.turs/) and */gar.gar.i̯ō/, any one of which could well be significant. First is the fact that the *r*-loss in μάρτῠς occurs in a final syllable, where phonological behavior is not necessarily comparable to treatments in medial position. Indeed, there is little consistency in the Greek treatment of final */-rs/: Cret. nom. sg. μαιτυρς shows final /-rs/ on the surface (probably restored, cf. dat. pl. μαιτυρσι), but the first /-r-/ has undergone dissimilatory glide formation; Aeol. μάρτυρ shows *s*-loss and retention of both *r*'s; and note further Dor. μάκαρς 'blessed', without *s*-loss in a final /-rs/ sequence, as against Att. μάκᾱρ, with *s*-loss and compensatory lengthening. Moreover, the consecutive syllables in */gar.gar.i̯ō/ are identical not only with respect to coda-final /-r/, but in all other respects as well — and indeed, this is part and parcel of the morphological category itself, which transparently shows "intensively reduplicated" forms of this type, much as in the Sanskrit intensives to which they are often compared. There is thus little reason to expect *r*-loss (or, if it occurred, to expect it not to be undone) in reduplicating forms of this kind.[6] In the end, it is not even clear that μάρτῠς is best interpreted as having arisen through dissimilatory *r*-loss at all. The case is clearly isolated, and should be viewed in that light.

3.4. In contrast, Lejeune's first type of *r*-loss (Arg., Arc. φᾶτρᾱ, Cypr. ϝρᾶτᾱ, etc.) is much better attested, suggesting that dissimilatory *r*-loss in Greek, even though clearly not obligatory, might in some sense be "regular", and that the unmarked version of the process might be precisely this one. Oddly, from our perspective, Lejeune himself omitted mention of ἀργός, a form that appears nowhere in his book. But this is probably without significance, since "French school" scholars regularly include the form in their presentations of the matter, which in all other respects tend to be based on Lejeune's

6 On *r*-retention in reduplicated forms (cf. ἀραρ-, ὀρωρ- above): Blanc 1999:320.

treatment.[7] It is appropriate, then, to record at this point two preliminary observations concerning ἀργός, based on the material presented so far. Note first that if ἀργός does indeed arise from dissimilatory *r*-loss, it would show *still a third* phonological environment, distinct from the two that appear in Lejeune's data, i.e. /$Vr_1.Cr_2V$-/ > [$Vr_1.C\emptyset V$-] (*r*-loss in an onset cluster directly after a coda-final /r/, if that is the correct interpretation of the syllable structure). Secondly, none of the data presented so far shows *r*-loss after a velar stop, as has allegedly occurred in ἀργός, a distributional fact that could in principle have phonological significance.

4.1. In order to pursue the ἀργός question, and to develop a better understanding of the phenomenon generally, the logical next step would be to gather more data about dissimilatory *r*-loss in Greek. It will come as no surprise to Hellenists that, first of all, the relatively scanty additional data were not easy to find,[8] and moreover that some forms have disputed etymologies or other attendant difficulties. A full discussion of these matters is impossible to undertake within the scope of an article, and the following survey lays no claim to exhaustiveness. Nevertheless, I believe that the picture sketched above can be filled out in a meaningful way that will allow for further analysis.

4.2. Let us first consider cases in which the *r*-less form and a corresponding form with /r/ are both attested.[9] Lejeune provided, of course, only a selection of the dialect material belonging to the word for 'phratry' (and related forms), which can easily be supplemented: thus, like the Arg./Arc./Phoc. φᾶτρ- forms already cited, also Arc. φᾶτριταις, Att. φᾶτριων, Cos (Διος) Φᾶτριῶ, PHETRIVM CIL XI 3614 (Naples, i.e. ~ Ion. φητριον); and similarly, like Cypr. ϝρᾶτᾶ, also Cypr. denom. *e-u-ve-re-ta-sa-tu/e-ve-re-ta-sa-tu* (ἐϝρᾶτασατυ).

In fact, forms belonging to the 'phratry' word can also undergo dissimilation of the type seen in Cypriot, with loss of the *second* /r/, as in φράτιον (Eust. 239.30; Tremblay 2003:131n61). Moreover, for Ion. φρηταρχος 'phratry-president' (Naples, 2nd c. AD), the question was already raised by Bechtel (3.91) as to whether the *third* /r/ was the "trigger" for the dissimilation.[10] Among items belonging to the same compound, a remarkable token[11] is Locr. φαταρχεω, where *both* /r/'s in the first member have undergone dissimilatory loss, and where there is little doubt that the third /r/ was indeed the trigger.

4.3. Several other items in which the *r*-form and an *r*-less form are both attested can be added to Lejeune's corpus, with varying degrees of certainty:

(i) Epid. ῥόπτον 'table sp. [?]' (IG IV^2(1).122.41 [4th c.] = DGE 109.41) and ῥοπτίον· κλειδίον (Hsch.) seem to belong with Att. ῥόπτρον 'wood in a trap, door-knocker, etc.'.

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7 See e.g. Blanc 1999:321, Le Feuvre 2005.

8 I am deeply grateful to Martin Peters for some help locating relevant bibliography.

9 For most of the material in this section and the next, see Poultney 1972:202ff. (largely based on Schmidt 1895:456f. and Wackernagel 1897:9); further Buck 1955:64 and relevant sections in Bechtel and Thumb-Kieckers.

10 It will be convenient, here and below, to make use of two terms from current phonological parlance, namely "trigger" (the conditioning element in a phonological process) and "target" (the segment that is affected by the phonological rule).

11 See Tremblay 2003:131n61, with further references.

(ii) The varia lectio θρεπτὰ or θρέπτα (Zenod. ad Δ 478 θρέπτρα 'cost of upbringing') is of the same general type.

(iii) The variation in the onomastic forms Cret. Ἐυρυστατω, Ἐυρυστατος ~ Ἐυρυστρατω/Ευρυσταρτω introduces a degree of uncertainty about the analysis (more on which below).

(iv) Wackernagel originally defended his dissimilated ἀργός (< *ἀργρός) on the basis of a well-established series of forms related to ὄρθρος 'daybreak', ὄρθριος 'early, in the morning': ὀρθαγορίσκος 'suckling pig' ~ ὀρθραγορίσκος (Ath. 4.139b, 140b; lit. 'marketed in the morning'),[12] ὀρθρογοή 'early-wailing' (Hes. *Op.* 568, vv.ll. ὀρθο-, ὀρθοβοή; see West ad loc.), the mythical dog Ὄρθρος (Hes. *Th.* 293, 309, 327, v.l. Ὄρθος; so West, see ad 293), and ὀρθολάλος 'early-twittering' (AP, for ὀρθρο-).

(v) A Dor. τᾱρόν· ταχύ is found beside τρᾱρόν· τ[ρ]αχύ, cf. τρη[ι]ρόν· ἐλαφρόν, δειλόν, ταχύ, πλοῖον μικρόν (all Hsch.) and Hom. τρήρων (2.1. above).

4.4. Naturally, the data for which only *r*-less forms are attested raise various difficulties, although at least several such forms seem plausible enough under a dissimilatory interpretation:

(i) If γαστήρ 'belly' [Hom.+] is correctly etymologized, as traditionally, via a root **gras-* (cf. γράω 'gnaw, eat', Skt. *gras-* 'feed on' and *grastar-* 'eclipser' [Hariv.] < *'swallower'), an original */gras-/ is assured (although the precise nature of the dissimilation process involved will concern us further below).

(ii) Similarly, χόνδρος 'granule; groats' [Hdt.+] is generally derived from a *χρόνδρος, cf. OE *grindan* 'grind', etc.

(iii) Less familiar is χειρόμακτρον 'hand-towel' [Hdt.+] < *-ομαρκτρον (cf. ὄμαρξον· ἀπόμαξον Hsch.), for which the connection with ὀμόργνυμι 'wipe' is surely attractive.

(iv) The epithet λάμπουρις (a designation of the fox) [Aesch.+] may derive from *λάμπρ-ουρ- 'shiny-tail' (cf. λαμπρός 'shining').

(v) The native etymology of χεδροπά 'leguminous fruits, pulse' [Hippoc.+] in terms of *χερ-δροπ- 'hand + pluck' may well be correct (cf. Viredaz 2000:294, 2002:140).

(vi) Although not mentioned in *EWAia,* Meier-Brügger's etymology of πέτρα/Ion. πέτρη, πέτρος 'rock' [Hom.+] < **per-tr-* (1980) provides a welcome connection with IE material elsewhere (cf. Ved. *párvata-* 'rocky' etc.).

(vii) It would be rash to insist on any particular account of the background of κροκόδιλος 'lizard; crocodile'; but the combination of κρόκη 'pebble, gravel' and δρῖλος 'worm' is widely accepted (see GEW s.v.).

(viii) Finally, as for καθαρός (Dor. κοθαρός, Aeol. κόθαρος) 'pure', καθαίρω 'purify' [Hom.+]: the connection with Ved. *śithirá-* 'loose', *śratharyáti* 'becomes slack' (Peters 1993:95ff., cf. *EWAia* II.636) is in many ways attractive (thus < ** $\hat{k}_{\mathring{r}}th_2ró$-, ** $\hat{k}_{\mathring{r}}th_2r$-*ié/ó*-).

4.5. The following problematic or more uncertain cases call for only brief comment:

[12] Perhaps Laconian, cf. Hsch. s.v. βορθαγορίσκεα.

(i) For νώθουρος· ὁ ἀδύνατος συγγίνεσθαι ἢ ὄνος (Hsch.), Wackernagel assumed a derivation based on νωθρός 'sluggish' (thus 'having a sluggish tail', i.e. 'tail' *sensu obscoeno*; for *νωθρ-, cf. νωθροκάρκιος· βραδὺς κατὰ λογισμόν Hsch.). But Lagercrantz (1896[1899]:275) may have been right to think that the basis is rather the *s*-stem adj. νωθής 'id.' [*Il.*+].

(ii) For Epid. θυρωτοιν 'doorways' (e.g. Collitz-Bechtel SGDI 3325.305), allegedly for (unattested) *θυρωτρ- (Buck 1955:361 s.v.), a simpler analysis as a substantivized participle θυρωτός 'with a door' (← θυρόω 'furnish with doors') is available (so LSJ), and may well be preferable.

(iii) Attic onomastic material of the type Ἀρχεστατος (cf. Cret. Ἐυρυστατω etc., 4.3.(iii)) belongs to a more general pattern: as Threatte points out (1980:480), similar *r*-less forms *without* /r/ trigger raise the possibility of a cluster reduction /-str-/ > /-st-/; and as often, simple error is difficult to exlude for some of the documents in question.

(iv) Forms like Att. inscr. θερμαστις (meaning uncertain) and θερμάστιον 'kettle' [Aen. Tact., inscr.] are often taken to show dissimilatory *r*-loss vis-à-vis θερμαστρίς 'id.' [Eupol.+], θερμάστρα 'oven' [Callim.+] (cf. θερμάζω 'heat').[13] But the suffix morphology is perhaps to be interpreted differently; see Threatte (1980:482f.) on "variant formative elements".

(v) The word χῖδρον [Alcm.], χῖδρα 'wheat-groats' [Aristoph.+] has no clear etymology, though Pisani (see GEW) has conjectured a preform *χριδρ-, comparing κριθή 'barley'. Even if the comparison is valid, the nature of the connection is unclear, and the phonetic relationship can be interpreted differently, e.g. via "*r*-interversion": cf. Tarent. τραφον for ταφρον 'ditch' and other such forms (Bechtel 2.395).

(vi) Hom. δένδρεον/Att. δένδρον 'tree' is sometimes given a preform *δρενδρ- (see Hackstein 2002:213, with references); but other (perhaps equally attractive) interpretations have been proposed, e.g. via a reduplicated *δερ-δρ-, a compound *δεμ-δρ-, etc. (see Blanc 1998:123).

(vii) According to a suggestion of Forssman (apud *EWAia* I.341), κήρυξ 'herald' and Ved. *kārú-* 'praise-singer' could go back to a */krāru-/ < **kreh₂-ru-*. But this is uncertain (see Hackstein 2002:213, and further discussion below).

(viii) Until recently, the *r*-less version of the word for 'fennel' (Att., Ion. etc. μάραθον, Sicil. μάραθα [Epicharm.], and toponyms) was taken to show dissimilatory *r*-loss vis-à-vis its Hellenistic variant μάραθρον (e.g. GEW II.173); but Myc. *ma-ra-tu-wo* suggests a different analysis (more on which below).

(ix) According to an intriguing suggestion of Dieter Gunkel (p.c.), καμάρα 'vaulted area' [Hdt.+] — a word that otherwise lacks a good etymology — could derive from a dissimilated *κραμαρ-, to the root of κρεμάννυμι 'hang'.

4.6. A few instances of dissimilatory *r*-loss have been suggested for Mycenaean, despite the indeterminacies of the writing system (or perhaps, in part, because of them):

(i) Under the assumption that Linear B sign **35* corresponds to a reading /pra-/,[14] Perpillou (1984:214ff. ~ 1996:146ff.) has suggested a gen. sg. /pʰrātós/ for /pʰrātrós/, in the

[13] So explicitly e.g. GEW I.665, Perpillou 1996:148.

[14] Far from an easy assumption, although the problem cannot be pursued here; for a different conception (**34/*35* = *ru₂* [lu]), see Hajnal 2006:34, with references.

phrase *wo-jo *35-to* /u̯ojo pʰrātos/ 'of his brother' (PY Eb 472). The suggestion is interesting, but I follow Tremblay (2003:30f., interpreting rather as /pʰrāto(r)s/) in considering it unlikely that such a form — if it ever existed — would not have been quickly restored in an inflectional context of that kind, as opposed to all other examples of dissimilatory *r*-loss, which involve either derivational patterns or alternations of root shape.

(ii) The anthroponyms *a-ko-da-mo* (TH) and a-*ko-ro-da-mo* (TH, KN) have been taken by Ruijgh (2003:224) to reflect /Argo-dāmos/ for the first and, with Wackernagel's long-lost **argró-*, /Argro-dāmos/ in the second. But even if the Thebes forms refer to the same person (which is highly uncertain), this seems very unlikely. A gen. sg. *a-ko-ro-da-mo-jo* was already attested at Knossos, and by far the most reasonable (and the most widely accepted) interpretation for *a-ko-ro-da-mo(-jo)* compares the actually-attested name Ἀκρόδημος.[15]

(iii) Much more attractive is Leukart's suggestion (1983:236) that the odd *t*-formant of Myc. *ke-ni-qe-te-we* /kʰernikʷtēu̯es/ 'hand-washing basins' (KN, MY) might have its origin in a */kʰernikʷton/, itself based on a dissimilated form of an original instrument noun */kʰernikʷtron/.[16] But once again, simple cluster reduction is difficult to rule out in such a case.

5.1. Two preliminary observations on the data are in order, before proceeding to a brief discussion of dissimilation in modern phonological theory.

To address, first, the question of the *non-application* of dissimilatory *r*-loss: even if the process is in some respects "regular" (cf. 3.4.), it is evidently not obligatory, as is true for most dissimilation rules. Moreover, such a process is likely to be blocked or undone under certain circumstances, such as within inflectional paradigms (as pointed out by Tremblay, cf. 4.6.(i)), or in order to maintain the shape of productive morphemes or transparent roots. This effect is evident if one considers, for example, the prefixes προ- or ὑπερ- preceding roots beginning with consonant plus /r/,[17] or *r*-final roots preceding the instrument suffixes -τρον/-θρον.[18]

Indeed, the oxytone adjective suffix -ρό-[19] could be considered in the same terms. There are several indexes of its productivity, or at least its synchronic transparency: note, for example, that even at a relatively late period, an oxytone adjective χονδρός 'coarse-grained (like groats)' [Hippoc.+] seems to have been created secondarily after the noun χόνδρος [Hdt.+] (4.4.(ii)); and one must assume that at earlier periods, adjectival -ρό- was even closer to being productive. Even on this basis, then, Wackernagel's claim about ἀργός cuts an unlikely figure: at the stage of Proto-Greek or earlier, we must envision a *ro*-stem adjective **argrós* developing to **argós*, where the blind application of dissimilatory *r*-loss would have had the unfortunate effect of rendering a transparent suffix opaque.

5.2. Secondly, it is apparent that forms like ἀργός or γαστήρ (4.4.(i)), with possible prehistoric applications of dissimilatory *r*-loss, differ in important respects from those forms in which some type of surface or synchronic variation is visible, whether this involves

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15 For discussion and further references see García Ramón 2009:4f. with n. 12.

16 Cf. τὰ νίπτρα 'wash-water' [Aesch.+], with the instrument noun a more likely basis for the Mycenaean form than a formation like νιπτήρ 'wash-basin' [NT] or νίπτης 'washer' [pap. VI AD].

17 E.g. προγράφω, προδραμεῖν/προτρέχω, προπράσσω, πρόφρων, etc.; ὑπερβριθής, ὑπερδραμεῖν, ὑπερφρονέω, etc.

18 E.g. δέρτρον, φέρτρον, ἄρθρον, τέρθρον, etc.

19 On this formant see recently Probert 2006 *in extenso*.

distinctions of dialect, or sociolect, or other factors affecting linguistic variation that are no longer recoverable. Still, it seems justifiable — partly for heuristic reasons, given the paucity of usable data — to operate *ex hypothesi* as if Greek speech, with respect to this phonological feature, was relatively uniform across time and space.

6.1. Despite the somewhat marginal status of dissimilation (and, even more so, dissimilatory loss) in historical phonology, dissimilation has played a more central role in the development of modern phonological theory. In certain respects, of course, the study of dissimilation was already well advanced before the 20th century, as in cross-linguistic studies of the typology of dissimilation, like those of Grammont (cited above, 3.1.). Yet even in this area, recent studies have brought new rigor and sophistication to the task. Thus, certain descriptive parameters of dissimilation have long been broadly familiar to historical linguists, such as the relative frequency of liquid dissimilation, or the distinction between "local" as opposed to "long-distance" dissimilation; but these somewhat crude observations have now been subjected to considerable fine tuning in thorough typological studies such as Suzuki 1998 (see the convenient summary in Alderete and Frisch 2007:382ff.).

6.2. In a similar way, Indo-Europeanists are familiar with the PIE root structure constraint that bans sequences with the shape **DeD-* — in effect, a kind of "similarity avoidance", a general principle that has loomed large in much theoretical work on phonology since the 1970's (from that time also referred to under the rubric of the "Obligatory Contour Principle" or OCP). The literature on this topic is now quite large, and its application to various types of phonological problems (in a variety of current frameworks) involves many technicalities; none of this is essential for present purposes.[20]

6.3. It is, however, useful to consider the reigning theoretical account of dissimilation, J. Ohala's "hypercorrection theory", proposed and developed in a series of articles, beginning in 1981 and now summarized in 2003:677ff.[21] A convenient paraphrase of Ohala's theory is given by Hale (2007:142n25): "'dissimilatory' sound change is frequently due to the hearer's correctly noting some acoustic feature present in a given segment, but mistakenly attributing the cause of the presence of that feature to 'spread' from an adjacent segment"; or similarly, as expressed by Hall (2007:1): "dissimilation stems from perceptual errors, when a listener hypercorrects for perceived assimilation"; or as Ohala himself puts it (2003:680): "it is listeners' *normalization* of what they imagine to be a distorted signal that leads to dissimilation". Ohala goes on to say, more generally (in the context of a discussion of Grassmann's Law), that "sound change — assimilative as well as dissimilative — is not related to physiological cost but is primarily a parsing error on the part of the listener". Thus for Ohala, a dissimilatory sound change like Grassmann's Law should not be ascribed to "ease of articulation", despite traditional claims to this effect.

6.4. One of the most attractive points of Ohala's theory is its ability to explain certain specific facts about the nature of dissimilatory change, such as which phonetic features tend to be subject to dissimilation. As Ohala notes (ibid.), this includes "features whose acoustic-perceptual cues are known to spread over relatively long time intervals", including aspiration (cf. Grassmann's Law), glottalization, retroflexion, palatalization, etc.

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20 For recent discussions of the OCP and similarity avoidance, see Frisch et al. 2004, Alderete and Frisch 2007:388ff. (with the earlier references on p. 389), Walter 2007. For a modern approach to PIE root structure constraints: Cooper 2008.

21 Note also the discussion in Alderete and Frisch 2007:384ff.

Even more closely related to our concerns here is a recent and very rich study of *r*-dissimilation by N. Hall (2007), mainly concerning dissimilatory *r*-loss in American English. In general, Hall's data and analyses largely support (and extend) Ohala's theory. It will be convenient to list here some of the major findings of her work that are most relevant to the following discussion of the Greek material:

(i) Dissimilatory *r*-loss in English reveals a number of non-random patterns, such as sensitivity to the nature of surrounding consonants.

(ii) Ohala's perceptual account predicts a pattern of "hypo-correction", i.e. *insertions* of /r/ that are the reverse of a given type of dissimilation. (This is particularly noteworthy, insofar as "liquid assimilation" is otherwise said to be very rare.)

(iii) Despite Ohala, not all types of *r*-dissimilation are necessarily perceptual in origin.

7.1. We may now return to the Greek data. I have already suggested (3.4.), based on a preliminary examination of Lejeune's material, that Greek dissimilatory *r*-loss may display certain regularities; and we may now confirm that the best-documented type shows consecutive syllables in which the /r/ is post-consonantal and part of the syllable onset, as in the following two subtypes:

(i) with *r*-loss in the onset of the first syllable: /$Cr_1V(C).Cr_2V$-/ > [$C\emptyset V(C).Cr_2V$-] (e.g. Arg., Arc. φᾶτρᾶ, Phoc. φᾶτριᾶ [2.1.], Cos Φᾶτριῶ, (Ion.) PHETRIVM [4.2.], etc.; χόνδρος [4.4.(ii)])

(ii) with *r*-loss in the onset of the second syllable: /$Cr_1V(C).Cr_2V$-/ > [$Cr_1V(C).C\emptyset V$-] (e.g. Cypr. *ve-re-ta,* δρύφακτος [2.1.], φράτιον, φρηταρχος [4.2.], θρέπτα [4.3.(ii)])

7.2. A form like Cypr. *e-u-ve-re-ta-sa-tu* (4.2.), in which the first of the two syllables in question is non-initial, is only trivially exceptional, as this is merely a denominative verbal form with augment. More interesting is κροκόδ(*ϱ)ιλος (4.4.(vii)), with non-consecutive syllables (assuming the traditional dissimilation analysis). Here it may be significant that the intervening syllable is identical to the first syllable (cf. "broken reduplication"), except for the lack of an /r/.

7.3. Although they approximate the schemes in 7.1., the following two items are somewhat deviant: Epid. ῥόπτον (4.3.(i)), apparently with /$r_1VC.Cr_2V$-/ as input (i.e., with a plain initial /r/ as trigger); and (ii) τᾶρόν (4.3.(v), for [*]τρᾶρόν), apparently with /$Cr_1V.r_2V$-/ as input (where the trigger is again a plain /r/, this time in the second syllable). These cases may benefit from consideration of the phonetics of Gk. /r/, though this is not a simple matter, and the problem cannot be examined here in detail. Broadly speaking, however, this was an apico-alveolar or apico-palatal trill (so e.g. Lejeune 1972:141f.).[22] But a number of factors serve to complicate the Greek situation: note, for example, the vexed question of the phonetic reality behind the rough breathing for initial ῥ-, or the phonetic interpretation of medial *r*-gemination (ῥέω 'flow' vs. εὐρρεής 'fair-flowing', etc.), and the probable voicelessness of /r/ following φ, θ, χ (see Allen 1987:43). Ultimately, the Greek data should be confronted with modern typological studies of the phonetics and phonology of "rhotics" — for example, the typology of rhotic duration contrast and neutralization (cf. the

[22] Allen (1987:41) prefers to assume an alveolar trill.

contrasting tap vs. trill in Spanish), as studied extensively in Bradley 2001a (and many published papers, e.g. 2001b). Thus, even though Greek had only a single /r/ phoneme, it may be that the two /r/'s in ῥόπτρον and τρᾱρόν were phonetically quite different; Bradley has shown, in any case, that a high degree of perceptual salience is characteristic of both initial position and intervocalic position for rhotic duration contrasts, which may help explain why dissimilation in ῥόπτρον and τρᾱρόν behaved according to the basic schemes in 7.1.

7.4. Of further interest is the double *r*-loss in Locr. φαταρχεω (4.2.), in an underlying triple-/r/ context. In her data-rich study, Hall (2007:9) reports only a single example of double *r*-loss from English, i.e. in *no(r)the(r)ner*.[23] The limited nature of the material renders analysis difficult; but one may conjecture that word length plays a role in such forms.[24] In this connection, note also Hall's discussion of long-distance /r/ resonances, and the primarily anticipatory nature of *r*-dissimilation (2007:18), the latter clearly a feature of this particular case.

7.5. There is, moreover, a potentially significant *segmental* regularity that emerges from the data discussed so far. In every case, at least one instance of post-consonantal /r/ in an onset syllable has a dental stop preceding the /r/:

(i) with *r*-loss in the first /Cr-/: / p^h(r)ātrV-/, /k^h(r)ondrV-/, /t(r)ārV-/
(ii) with *r*-loss in the second /Cr-/: /u̯rēt(r)V-/, /drup^h(r)V-/, /p^hrāt(r)V-/, /t^hrept(r)V-/, /krokod(r)V-/

In other words, there are no cases of dissimilatory *r*-loss in one of two consecutive onset syllables unless one of those syllables has a post-dental /r/ (thus no examples of the types /p^h(r)VkrV-/, /krVp(r)V-/, /pVk^h(r)V-/, /k(r)Vk^hrV-/, etc.). To this it might be countered that such input forms are rare in the language anyway (apart from items likely to be resistant to *r*-dissimilation [5.1.]), and that /t/ is common as a formant. The first point may or may not be valid; but in response to the second point, one might note that the dental in question is often /d/ (highly rare as a morphological formant) or often not a formant (or even part of a formant) at all, as in /k^h(r)ondrV-/, /t(r)ārV-/, /drup^h(r)V-/, /t^h-/ in /t^hrept(r)V-/, /krokod(r)V-/. Moreover, similar consecutive-syllable /$Cr_1V.Cr_2V$-/ patterns are also found in Hall's data from American English: e.g. *frat(r)icide, Trist(r)am Shandy, imp(r)opriety, f(r)ustrated/f(r)ustration, spect(r)ogram, inf(r)astructure.* Here, in fact, there is a significant correlation with */labial/* consonants, a point to which we will return. Hall has also found that deletion of an onset /r/ (as in these English examples) is more likely to be triggered by another onset /r/ than by a coda /r/, which appears to be true of Greek as well.[25]

8.1. Many of the remaining Greek examples fall into a different descriptive pattern, where once again (as in e.g. Epid. ῥόπτον) plain /r/ (and not /Cr/) acts as trigger,

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[23] Note that this is an American English form, i.e. unrelated to the *r*-loss that would be normal for various forms of British Received Pronunciation. Hall also notes (ibid.) that no examples of dissimilation affecting two out of three liquids were reported by Walsh Dickey (1997).

[24] In a more broadly Indo-European context, note e.g. Balles 1997 on this factor.

[25] Hall suggests (p.c.) that this type of case is *not* likely to involve perceptual error, since many examples involve the onsets of stressed syllables, where perceptual mistakes are unlikely; thus, *pace* Ohala (6.3.), this is perhaps rather a matter of articulatory difficulty.

but with a different profile, including the following features: there is a three-syllable minimum; the /r/ is non-initial and may precede or follow the target syllable; the target is in an onset syllable with post-consonantal /r/, including the possibility of a complex onset with initial /s/ (thus /CrV/ or /sCrV/); and a syllable may intervene. Hence the following instances (or possible ones, since some of the data are insecure, as discussed above):

(i) /r/ precedes /(s)Cr/: Εὐρυστ(ϱ)α- (4.3.(iii)), θυρωτ(*ϱ)ο- (4.5.(ii))
(ii) /r/ follows /Cr/: λαμπ(*ϱ)ουρι- (4.4.(iv)), νωθ(*ϱ)ουρο- (4.5.(i))
(iii) /C(r)V.CV.rV/ (with intervening syllable): ϰ(*ϱ)αθαρο- (4.4.(viii)), ϰ(*ϱ)αμάρα (4.5.(ix))

8.2. It is interesting to compare Lejeune's list of well-formed double-*r* words (2.1.). Thus, a three-syllable form like βάρβαρος does not match the profile just described, unlike (for example) Att. βάραθρον 'gulf' or ἐρυθρός 'red', both of which would in principle be susceptible to *r*-loss, if this were not likely to be blocked by the salience of the suffixes (5.1.).

8.3. It is impossible to say whether the frequency of dental stops in this type (insofar as it can really be isolated as a type) carries any significance, given the limited nature of the data and the uncertainties surrounding some of the forms in question.

9.1. We must next consider cases in which the target or trigger appears in the coda position of a syllable. To return for a moment to Lejeune's example μάρτῦς (2.1.), based on a form with /r/ in the codas of consecutive syllables: I suggested (3.3.) that the case is in fact isolated, and that it is difficult to know exactly what has accounted for the *r*-loss in this form, which could in theory be attributed to other deletion processes (including "absorption", cf. Att. μάκᾶρ). To be sure, the data from Hall's study (2007) indicate that *r*-loss in coda position is more likely to be triggered by another coda /r/ than an onset /r/ (cf. the converse regularity for onset /r/'s [7.5.]); yet it is normally the *first* /r/ that tends to be lost in these situations, and not the second (as would be the case in μάρτῦς, under a dissimilation analysis).

9.2. Nor should we be detained at this point by several other coda /r/ forms that pose difficulties of various kinds. In χειρόμα(*ϱ)ϰτρον (4.4.(iii)), we have a triple-/r/ context with each /r/ in a different syllabic position, an unclear trigger, and a complex cluster involving the transparent instrument suffix -τρον. In φρητ(ϱ)αρχος (4.2.), we have another triple-/r/ case with unclear trigger. And as for ϰαθαρός (4.4.(viii)), accepting Peters' etymology: there are serious indeterminacies about the chronological stage at which the dissimilation occurred, which in turn affect our understanding of the phonological context on which the process operated: e.g. Pr.-Gk. **kərtʰaró-* or **kərtʰərá-* or even earlier (cf. Ved. *śithirá-*), or at the stage of later dialectal versions like pre-AI **kratʰaró-*, etc.

9.3. More interesting are the remaining few coda /r/ cases, which are remarkably uniform:

(i) /V(r).CrV/, C = dental: χε(*ϱ)δρο- (4.4.(v)), πε(*ϱ)τρα/ο- (4.4.(vi))
(ii) /Vr.C(r)V/, C = dental: ὀρθ(ϱ)α-/ο- (4.3.(iv))

In each of these cases (provided, for the items in (i), one accepts the etymologies discussed above), an /r/ in syllable-final position is directly followed by a post-consonantal /r/ in syllable onset position, and the consonant is always a dental stop. Note that in Hall's

(otherwise very extensive) data, this type is also rare, seen in words like *su(r)prise, interp(r)et, ente(r)prise,* and a few more; and they are mainly restricted to labial contexts, otherwise an important feature in her data generally, as mentioned above (7.5.). This correlation suggests that the dental restriction just noted (and posited for other Greek data above, especially the consecutive /Cr-/ onset type [see again 7.5.]) may also be significant.

10.1. We are now in a position to explore some possible applications of the above material to particular forms, concluding with final remarks on the problem of Gk. ἀργός, with which we began.

For γαστήρ (4.4.(i)), I believe that the background of the dissimilation can be clarified, and that, as a result, residual doubts about the etymology can be laid to rest.[26] In fact, the dissimilation context for this word is loosely treated in the handbooks, most of which give a preform *γραστήρ as the explicit or implied basis of the dissimilation.[27] But this traditional *γραστήρ corresponds as such, as we have seen, to no documentable pattern of Greek dissimilatory *r*-loss, and thus a nom. sg. *γραστήρ itself is not likely to have been a serious factor in the process. Much more likely is the consecutive-onset-syllable format *γραστρ- (so cited by Meier-Brügger 1992:II.100), available in at least two sources: first, the original oblique **gras-tr-V́* to hysterokinetic **gras-tḗr* (cf. Hom. dat. γαστρί 3x *Il.*, gen. γαστρός 1x *Od.*); and secondly, early derivatives with the shape */grastrV-/, their descendants already well-established in Homer (cf. γάστρη 'paunch [of a vessel]' 1x *Il.*, 1x *Od.* and ὁμογάστριος 'from the same womb' 2x *Il.*). As an alternative (or an additional input): even the remodeled declension with obl. **gras-tér-os* etc.[28] corresponds formally to the "3-syllable minimum" type described above (8.1.).

10.2. The archaic *u*-stem adjective Hom. κρατύς 'strong', beside the *s*-stem Caland abstract Hom. κράτος (~ κάρτος, Aeol. κρέτος) 'strength', is largely replaced (beginning already in Homer) by a *ro*-adjective. But that form has the shape κρατερός/καρτερός, and not (as one might expect) †κρατρός/†καρτρός. Now the replacement of **-ró-* by Gk. **-eró-* is a general problem,[29] and more than one factor may be involved in any individual instance. Here, for example, it may be relevant that both /kra.tro-/ and /kar.tro-/ correspond to two of the most highly-favored contexts for dissimilatory *r*-loss, i.e. the /$Cr_1V.Cr_2V$-/ type for the first (including presence of a dental consonant in at least one /Cr-/ sequence [7.5.]), and the /$Vr_1.Cr_2V$-/ type for the second (including a dental consonant [9.3.]). Together with the synchronic stability of the full stem-forms /krat-/ and /kart-/ (cf. the *u*-stem and the *s*-stem), and given the undesirability of eliminating /-ro-/ as (at least part of) a productive or transparent formant (5.1.), the stage is set for a replacement by **-eró-* (i.e., as an alternative to dissimilation outcomes like †κατρός or †καρτός). Similar considerations might be applicable to other cases, such as (Caland) βρῖθος 'weight' [Eur.+], Hom. βριθύς 'heavy', but *ro*-adjective Hom. βριαρός[30] (i.e., not †βριθρός or any of its conceivable dissimilation outcomes).

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26 For these doubts: note Frisk's insistence about the etymology ("Ohne Zweifel ..."), in reaction to "[a]ndere Deutungen" and earlier resistance to the idea (GEW s.v.).

27 So e.g. GEW, DELG, Buck 1949:253, IEW 404.

28 The majority type in Homer: gen. γαστέρος, dat. γαστέρι, acc. γαστέρα, nom. pl. γαστέρες.

29 See García Ramón 1992:199 on the einzelsprachlich nature of Caland **-eró-* replacements like κρατερός/καρτερός.

30 Note also Hom. ὄβριμος 'heavy' as another substitute adjective in the system.

10.3. As already noted (4.5.(viii)), Mycenaean now provides an interesting perspective on μάραθον/μάραθρον, on which Hall's analyses shed further light. Myc. *ma-ra-tu-wo* now makes clear that the earlier first-millennium form μάραθον continues *μάραθϝον, and that the later version μάραθρον — as correctly seen by Chantraine (DELG) — is a hypercorrect (or, in Hall's terminology [6.4.(ii)], "hypo-correct") form.[31] We can add more specifically that μάραθον ~ μάραθρον corresponds precisely to alternations that would arise from the "three-syllable minimum" type of dissimilatory *r*-loss identified above (8.1.).

10.4. The first member of Myc. *re-u-ko-ro-o-pu$_2$-ru* (PY Jn 415, MN) (i.e. /leukro-op^{h}rūs/ or /leukrŏp^{h}rūs/ 'white-browed') incorporates a well-known problem, even apart from the apparently uncontracted composition vowel (on which see Meissner and Tribulato 2002:322): does the apparent *re-u-ko-ro-* (i.e. /leukro-/) constitute a real form, instead of the expected /leukó-/ (cf. λευκός 'white')? Most often, this is taken simply as an error for *re-u-ko-*.[32] In contrast, Hajnal (1995:137n173) apparently took the form as real: he envisioned an "etymological spelling" for contracted *<*re-u-ko-ro-pu$_2$-ru*>, but offered no explanation for **leukro-* itself. In an earlier work, I tentatively suggested (Vine 2002:332) that **leukro-* could perhaps be accounted for on the basis of a contamination, comparing other color terms like **erut^{h}ro-* 'red' (Myc. *e-ru-to-ro, e-ru-ta-ra*), **k^{h}lōro-* 'green', and possibly even a **lukro-* 'white' or 'shining' (if such a form existed in early Greek, i.e. as a counterpart to Ved. *rucirá-*). But it now seems more attractive to interpret this /leukro-/ as another "hypo-correct" form with secondary *r*-insertion, with a basis in the /Cr$_1$V.Cr$_2$V-/ consecutive-onset dissimilation pattern: i.e., /leukrŏp^{h}rūs/ (with its adventitious /r/) would have arisen in much the same way as μάραθρον (10.3.).

10.5. Hackstein (2002:213) was properly cautious about Forssman's suggestion of a dissimilated **kāru-* < **krāru-* (for both κήρυξ and Ved. *kārú-* [4.5.(vii)]), citing Hom. τρήρων 'fearful' — a form Lejeune had already used in a similar way (2.1.). Yet this word itself provides a probable case of exactly the sort Forssman suggests, i.e. τᾱρόν (beside τρᾱρόν), 4.3.(v); thus at least κήρυξ < **krāru-* remains possible.

Gk. τρήρων/τ(ρ)ᾱρόν is also interesting in a different way, given its historical basis in a form **tras-ro-* < **tr̥s-ro-* (i.e., with the zero grade of **tres-*: cf. τρέω 'tremble', aor. τρέσσαι, verbal adjective ἄτρεστος 'fearless'). Thus the preform */trasro-/ recalls the consecutive-onset /Cr$_1$V.Cr$_2$V-/ context, which leads to the question of what governed the phonological treatment of a sequence like */trasro-/. As always, more than one factor may have been operative: one thinks immediately of the probable synchronic relationship (at a time when a */trasro-/ was still intact) between the root allomorphs /tras-/ and /tres-/, as well as the transparency of the suffix **-ró-*, already mentioned on several occasions above. But a critical question concerns the syllable structure of */trasro-/: given the fact that at least historically, */sro-/ (more generally */srV-/) was a possible initial sequence, this might seem to suggest a syllabification */tra.sro-/; yet this is not likely to be correct. As can be seen in τ(ρ)ᾱρόν, **trasro-* underwent (so-called) "compensatory lengthening" to **trāro-*, and it is now clear that compensatory lengthening is sensitive to the nature of the lost segment and its syllabic position (De Chene and Anderson 1979): for the sequence in question, the deleted consonant must be in coda position for compensatory lengthening to

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[31] As for GEW: Frisk duly cites *ma-ra-tu-wo* in his *Nachträge*, but (as is typical of that work's lack of real engagement with the Mycenaean data) entirely without comment.

[32] Thus e.g. Docs.^{2} s.v. (following Palmer) and Lejeune (1972:143).

occur, suggesting that the syllabification was in fact */tras.ro-/ — i.e., a sequence that does *not* show the consecutive-onset /$Cr_1V.Cr_2V$-/ context, and which thus would have been unlikely to undergo dissimilatory *r*-loss.

This material, in turn, offers an interesting contrast with what is probably the clearest case of dissimilatory *r*-loss within PIE itself, i.e. **tri-sr-(es)* 'three [f.]' > **tisr-* (Ved. *tisrás,* Av. *tišrō,* OIr. *teoir*); this may suggest, then, that for the stage of PIE in question, we should indeed assume a syllabification */tri.srV-/.[33]

10.6. We can now return to ἀργός, and some concluding observations about Wackernagel's celebrated "*ἀργρός > ἀργός" hypothesis. Note, first of all, that the dissimilatory *r*-loss affecting sequences of the type /Vr.C(r)V-/ is highly restricted, not only in Greek, but also in English (9.3.), where we now have very full documentation about dissimilatory *r*-loss (thanks to Hall 2007). Insofar as such forms are found in Greek at all, this type (and the related /V(r).CrV-/ type) are restricted to dental consonants, a context that may be significant (cf. the labial restriction visible for such patterns in English). We have seen, secondly, that, notwithstanding the existence of rare cases like ὀρθ(ρ)o- (4.3.(iv)), the salience of adjectival **-ró-* renders a development *ἀργρός > ἀργός at least somewhat surprising in any case (as opposed, say, to a development like †ἀργερός, cf. κρατερός/καρτερός [10.2.]). Moreover, an attractive recent analysis of Greek names in -ώ (Remmer 2007:129f.) suggests that ἀργός itself (cf. Ἀργώ) must have been available at a considerably early stage, so as to serve as part of the model for the creation of names like Γοργώ [*Il.*+], based on γοργός 'terrible'.

All of this leads to a kind of chronological conundrum: ἀργός is a Greek *unicum* (as a putative dissimilation result), and predates Homer, suggesting that a development like the one supposed by Wackernagel — if it ever *did* occur — is not likely to have taken place during the history of Greek; yet its supposed *ro*-stem cognates Ved. *ṛjrá-,* Av. *ərəzra-* (1.1.) show no sign of dissimilation. Nor is this a dissimilation type that is documented for PIE (apart from still-controversial instances involving dentals, cf. n. 33).

11. In the end, many may prefer to maintain their belief in Wackernagel's *ἀργρός. After all, one is free to imagine that an isolated dissimilation "event" of roughly the ὀρθ(ρ)o- type could have occurred at any sufficiently early stage between PIE and (approximately) the Mycenaean period. But barring such a random occurrence, *ἀργρός (which, of course, is anachronistically written in that form) must be considered an isolated and unverifiable relic from the remotest period of Greek linguistic prehistory, and one that is fundamentally unconnected with attested patterns of Greek dissimilatory *r*-loss. It has been possible to show — thanks in part to insights about dissimilation from modern phonological theory — that Greek dissimilatory *r*-loss displays a high degree of regularity, which in turn may help clarify a range of issues in the history of Greek, whatever one chooses ultimately to believe about "ἀργός < *ἀργρός".

[33] Cf. also Pinault (2007:275) on IIr. **strī-* 'woman' < **s(ŗ)-tr-ih₂,* "with regular dissimilation of the first liquid by the second" (more reminiscent of the Greek type presented in 9.3.(i) above). Differently on both 'three [f.]' and 'woman': Kim 2005, 2008, and forthcoming.

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